Question:
Carlos is planting a lemon tree. The tree will cost $90 to plant. Each year it will grow 7 lemons, which he can sell for $1.5 each. It costs $3 a year to water and feed the tree. How many years will it take before he starts earning money on the lemon tree?

Reasoning + Answer:

We are given:

- Cost to plant the tree: $90 (one-time cost)
- Each year, the tree produces 7 lemons
- Each lemon sells for $1.50
- Annual cost to water and feed: $3

We want to find out **after how many years** Carlos starts **earning money** âĢĶ that is, when the income from selling lemons exceeds the costs of planting and maintaining the tree.

LetâĢĻs define:

- Let y be the number of years after which Carlos starts earning money.
- Annual income from lemons: $7 \times 1.5 = 10.5$ dollars
- Annual expenses: $3
- So, net income per year after the initial planting cost: $10.5 - 3 = 7.5$ dollars per year

But since the initial $90 cost is incurred upfront, we need to find when the **total earnings (net of expenses) over y years** covers the initial $90 cost.

So, we set up the equation:

$$
\text{Total net income over } y \text{ years} = 7.5y
$$

We want this to equal or exceed the initial cost:

$$
7.5y \geq 90
$$

Solve for y:

$$
y \geq \frac{90}{7.5} = 12
$$

So, after **12 years**, the total earnings cover the initial cost. Therefore, **after the 12th year**, Carlos starts earning money (since in the 13th year, he is making a profit beyond the cost). But the question is: *When does he start earning money*?

This means the **first year in which he makes a profit**, i.e., when cumulative earnings exceed the initial cost.

After 12 years, the total net profit is exactly zero (he breaks even). So in year 13, he starts making a *profit* (since revenue exceeds expenses, and the cost has been covered).

But the question says: *"How many years will it take before he starts earning money on the lemon tree?"*

It might be interpreted as: after how many years of operation does he begin to make **profit**? So, **at year 13**?

Wait âĢĶ but actually, if we interpret this question as when he **starts earning money** (i.e., profits begin to happen), then this is when the **net income first turns positive**.

So at year 1, he incurs $90 upfront, then makes $7.5 per year after.

So **after year 12**, the net cumulative income is $7.5 \times 12 = 90$, which **covers** the initial cost. At that point, the **profit is zero** âĢĶ he has not yet started *earning* money (he hasn't made a profit, just broken even).

So he **starts earning money** only after that, i.e., **after the 12th year** âĢĶ so in the **13th year**.

Wait âĢĶ but some interpretations of "start earning money" could mean when the cumulative income exceeds the planting cost. So **after 12 years**, so it takes 12 years to break even.

Therefore, he **starts earning money** after the initial cost is recovered.

Thus, **after 12 years**, he no longer loses money âĢĶ at year 13, he earns profit.

So when does he **start earning money**?

Answer: after **12 years** âĢĶ when the cost is recovered.

So, he starts earning money after **12 years**, meaning **it takes 12 years** to break even.

So it takes 12 years to start earning money.

LetâĢĻs check:

- After 1 year: $7.5 \times 1 = 7